

Akash Gogate

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Education

University of Wisconsin-Madison

Graduating May 2028

Bachelor of Science in Computer Science & Biology — Dean's List Fall 2024, Spring 2025, Fall 2025 — 3.8 GPA

Technical Skills

Languages: Java, Python, Rust, TypeScript, C++, R, SQL, Linux, Bash

Systems & Tools: PyTorch, TensorFlow, Scikit-learn, Kafka, Docker, Kubernetes, MongoDB, DuckDB, FastAPI, REST APIs, GitHub Actions, Neo4j, OpenCV, Weights & Biases

Experience

Leidos

May 2025 – Present

Software Engineer Intern

- Built **Neo4j** knowledge graph with LLM plugin: **300–500** nodes, **1,000–2,500** edges for **30-engineer** team onboarding.
- Built **Kafka+MongoDB** pipeline at thousands of events/sec with dead-letter queue recovery.
- Automated regression testing via **Docker & GitHub Actions** across 5 system components.
- Eliminated **200+** satellite scheduling conflicts via **Python** memoization & DP across **1,000+** active satellites.
- Implemented **Rust** microservice for satellite scheduling validation at Leidos.
- Achieved **~95%** unit & integration test coverage via **Docker & Kubernetes** across **5** system components.
- Built **DuckDB+Redis** analytics layer with LLM natural-language query interface at **<1s** over **1M+** telemetry records.

The Kendzierski Lab at UW-Madison

Sep 2025 – Present

Machine Learning Research Intern

- Reduced scRNA-seq pipeline runtime **80→27 min** via **PyTorch** DataLoader optimization across **4** architectures.
- Cut manual literature review **~60%** via NLP entity extraction across **500+** papers.
- Automated experiment reproducibility via deterministic seeds & versioned checkpoints.

Inspirit AI Research Internship

Sep 2022 – Mar 2023

Machine Learning Research Intern

- Achieved **95%** skin cancer accuracy via **Decision Tree** on **70K** dermoscopic images using **scikit-learn**.
- Trained **Random Forest** on miRNA sequences achieving **95%** accuracy ($p < 0.05$) via stratified k-fold CV in **R**.

Projects

Self-Improving LLM Agent | [Github Link](#)

Jan 2026 – May 2026

- Architected failure-driven self-improvement loop outperforming ReAct **15pp** on 5,000+ tasks via LLM-as-judge.
- Built unified LLM serving layer across **Claude, Grok, Llama & Qwen** with **GitHub Actions** regression benchmarks.
- Built LLM-as-judge harness scoring **7** quality dimensions with failure-rate convergence tracking.

Lotus Health | [Github Link](#)

March 2026

- Built ICD-10 risk engine on 1,080-node graph over **45M** records via **Groq Llama** RAG.
- Designed hybrid **Groq Llama** / Python ICD-10 scoring for reproducible clinical outputs.
- Integrated **Deepgram** voice pipeline for real-time spoken-query ICD-10 risk scoring over **45M** records.

LLM-Based SAT Practice Test Generator | [Github Link](#)

Jun 2024

- Built full-stack SAT prep app in **TypeScript & FastAPI** with **GPT-4o** adaptive question generation & scoring.
- Designed REST API in **FastAPI** serving **75** cached SAT questions across **3** sections by user performance history.

Ecological Conservation Game | [Github Link](#)

Jan 2024

- Engineered **C++** simulation in **Raylib** sustaining **60+ FPS** across **10,000+** agents via $O(N)$ spatial hashing.
- Engineered multi-generational **4-species** evolution across **10,000+** agents via stochastic fitness mutation.

Leadership

TEL: Spatial Intelligence Ideathon

July 2026

- Won **2nd place** among **50+** teams at TEL Ideathon via **ZaiNAr** sub-meter positioning & **AR** wayfinding.
- Led **2-person** team building dual-product airport platform with **ZaiNAr** tracking for airline ops & wayfinding.

Princeton Racket Club

May 2024 – Aug 2024

Tennis Coach & Tournament Director

- Directed **8** regional tournaments at **380+** match entries with 100% on-time rate as dual director & head coach.
- Coached **8–10** athletes per class in **3** cohorts, advancing **3 juniors 150+** ranking pts & **5 adults 0.5 NTRP**.

Tennis Racket Stringing Services

Jan 2019 – Present

Founder & Owner

- Founded racket stringing business scaling to **45+** clients via team referrals, servicing **12–20** rackets/month.
- Maintained **3-day** standard and **2-day** premium turnaround SLA across competitive and recreational athlete clients.